DUNE Offline Computing Model Calculations for 2026

K. Herner, M. Kirby, and H. Schellman for the Computing Consortium

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1 Introduction

This is a projection for DUNE CPU and storage needs intended for use at the Computing Contributions Board meeting in December 2025. It projects needs for 2026 to 2030.

The overall computing model and 2023 projections for DUNE were described in chapters 6-13 of the (Oct. 2022) DUNE Conceptual Design Report [**?**]. This document provides updates on resource needs for 2026 - 2030.

The 2026 projection is done using codes at: https://github.com/DUNE/CCB-data/tree/CCB-Jun24/models-2024/ from parameters stored in a json and csv file. We use CPU and storage sizes derived from ProtoDUNE, 2x2 Demonstrator, and simulation experience and apply them to projected numbers of events from the various DUNE detectors.

Changes since the last report [**?**, **?**, **?**] include:

- A complete rewrite of the model code to make it more robust and allow more flexibility in plots.
- Delay and subsequent extension of the start of ProtoDUNE-VD running at CERN to happen 2025-2026. This increased and extended the need for storage/CPU into mid 2027.
- More data from both ProtoDUNE-HD and ProtoDUNE-VD than originally projected because of additional cosmic-ray operations and enhanced performance of the DAQ.
- Anticipate need to maintain both ProtoDUNE-HD and ProtoDUNE-VD raw data on disk simultaneously for reprocessing. The previous model assumed 1 year lifetimes for raw data on disk and this has been extended to 3 years. **This leads to a significant increase in the 2025 need which may need to be mitigated by reducing disk access to legacy ProtoDUNE-SP/DP data. See Figures ??, ??, ?? for projections.**
- This report now uses existing data catalog information up to August 2024 in forming a baseline for future projections.
- Switch to HS23-Years as opposed to Core-Years as the main CPU reporting system.
- Revisions to near-term requests based on the 2023 experience including a hold on tape requests from the collaboration during ProtoDUNE activities.
- Enhanced model for splits (Tables **??**, **??**, **??**) between regions in resource provision.

2 Model Summary

Resource requests are based on a processing and storage model that separates different phases of the detectors and includes 5 classifications of activity.

- Physics Data - described by number of events, CPU time/event, storage/event. Physics Data are described by number of events and then transformed into storage and CPU based on measured or estimated performance.
- Simulation - described by number of events, CPU time/event, storage/event. Currently both simulation and reconstruction are combined as a single entity as that is how we currently perform simulation. The model will be extended to support multiple stages in future.
- Test - data taking as part of commissioning. These have a short lifetime and are initially described by PB of storage. They are assumed not to consume large amounts of CPU and to only reside on disk briefly.
- Trigger Primitives (TP) - these are currently described by PB of storage and are assumed not to consume large amounts of CPU.
- Analysis - Analysis is assumed to be use little disk but consume considerable CPU and IO resources. It is currently described by a scaling factor ($\sim 50\%$) relative to Reconstruction and Simulation CPU.

Table **??** summarizes the main detectors and their abbreviations. Smaller prototypes that generate less than 1 PB of data are not explicitly included.

Abbrev.	Detector	Running time
SP	ProtoDUNE Single Phase	2018-2021
DP	ProtoDUNE Dual Phase	2018-2021
Coldbox	Both VD and HD	2018-2024
PDHD	ProtoDUNE-2 Horizonal Drift	2024-2026
PDVD	ProtoDUNE-2 Vertical Drift	2024-2026
2x2	Prototype NDLAr Detector	2024-2026
HD	Far Detector Horizontal Drift	2028-
VD	Far Detector Vertical Drift	2028-
NDLAr + TMS	Near Detector Liquid Argon + Muon System	2030-
SAND	On Axis Near Detector	2030-

Table 1: Detector abbreviations and estimated running times. Simulation campaigns occur earlier.

These inputs are then used to calculate storage and CPU needs.

2.1 Storage replication and retention policies

Disk storage policies are designed to optimize access and minimize CPU inefficiencies due to network transfer speeds. In particular, all recent data samples are planned to have at least one disk copy to avoid the need to access tape. As raw data processing is not I/O bound, one disk copy may be sufficient. Analysis of reconstructed samples is I/O bound so multiple copies located closer to CPU are desirable.

Each type of data has a storage retention policy which includes lifetimes on disk and tape and number of copies on disk and tape. For example, raw data has a very long tape retention policy and 2 tape copies, with a 2-3 year stay of 1 copy on disk. Recent simulated and reconstructed data samples typically have 1 tape copy, 2 disk copies and disk retention times of 1.5-2 years to allow fast analysis. An extended retention time after the end of data taking for the last version of sim/reco is added to allow for extended data analysis.

Simulation and reconstruction from 2018 ProtoDUNE run is still active but is reduced to 1 disk copy.

The model currently does not include any disk headroom allowance for data movement. This will be added in the 2025 version of the model.

2.2 Processing calculations

We assume one processing campaign per year, with all real data reprocessed each campaign. Simulation is assumed to be regenerated once per year.

CPU calculations are now made in HS23 units. Calculations for the ProtoDUNEs and DUNE far detector are based on existing reconstruction and simulation experience. We have much less experience with Near Detector codes so those estimates are less accurate.

We assume ProtoDUNE running in 2024-2025 with startup of DUNE FD commissioning in 2027 (Test stream) and data taking in 2029. A cap of 30 PB/year for raw data (Events, Trigger Primitives and Tests) is imposed by scaling all categories down proportionally when 30 PB is reached.

3 Details of the model

The model is implemented in python. Time series are created for each combination of detector, data type, resource and location. Those time series are then scaled, extended and iterated and the results stored as new time series. The end result is a csv file with a line for each permutation of types and figures summarizing various aggregations of data. Figure **??** shows the steps applied to data and simulation.

3.1 Inputs

Inputs to the model include projections of events and storage per year by detector and datatype, which are contained in a "timeline file". A master configuration file in json format is used to store detector

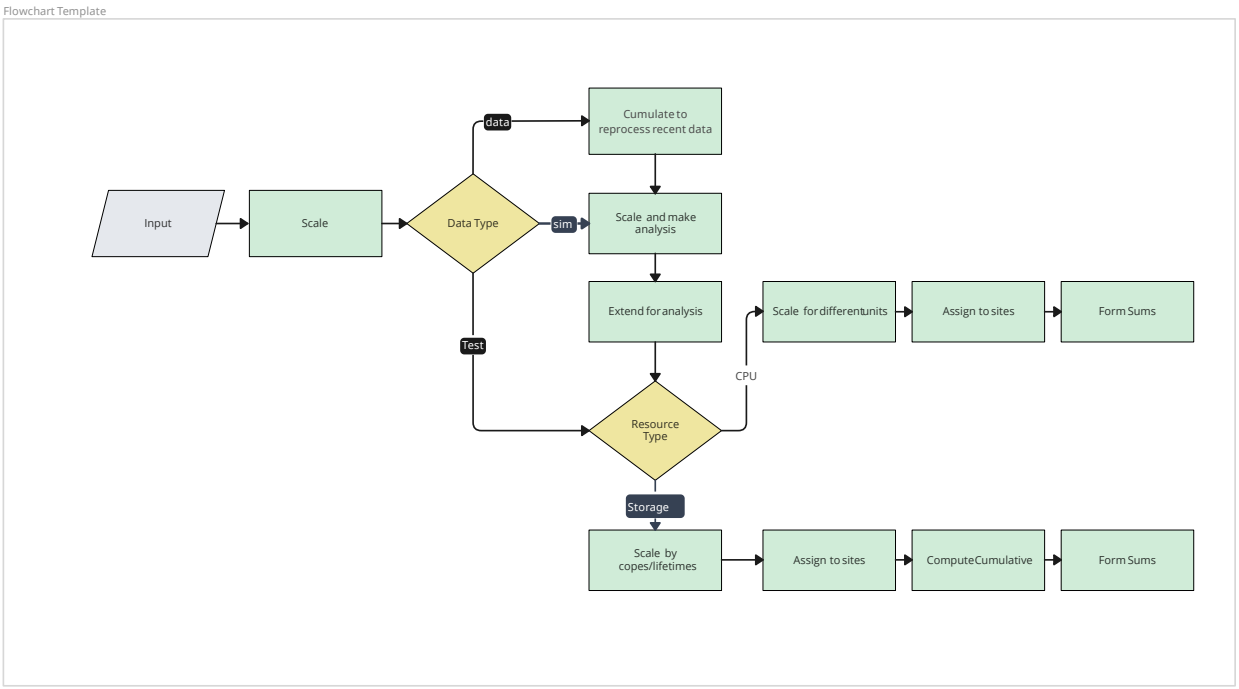


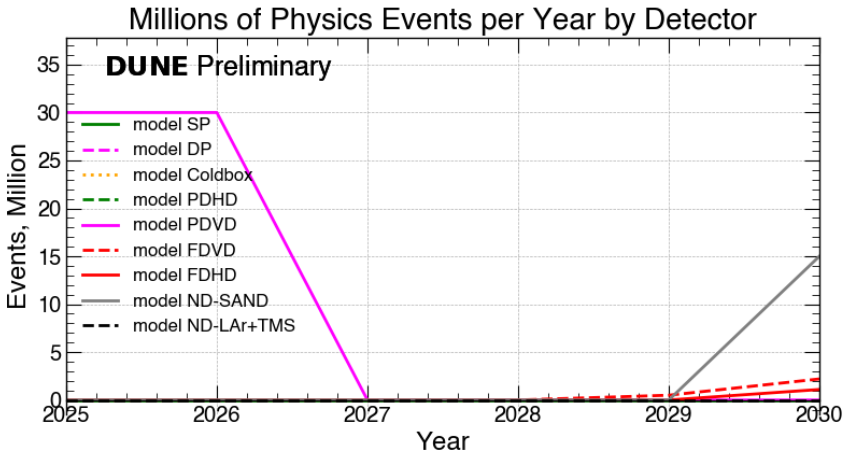
Figure 1: Flow chart of transformations used to transform raw event counts into summaries such as cumulative disk on tape.

specific parameters which do not depend on year.

The parameters are stored in NearTerm_2024-09-16-2040.json and NearTerm_2024-09-16-2040_timeline.csv

The calculation starts with numbers of events from the timeline[1] and uses those numbers, with CPU/storage times/event from Table **??** to calculate the annual new storage and CPU use for data taken in that year. Figure/Tables **??** and **??** show the assumptions for numbers of events/year.

[1]Event numbers for previous years are back-estimated from the total size of the data from the data catalog and an estimated size/event because the catalog does not report event number totals. Early simulation stored very large events relative to the current estimated size, leading to an overestimate of the number of events but a correct estimate of disk and tape use.



Detectors	2025	2026	2027	2028	2029	2030
SP	0.0	0.0	0.0	0.0	0.0	0.0
DP	0.0	0.0	0.0	0.0	0.0	0.0
Coldbox	0.0	0.0	0.0	0.0	0.0	0.0
PDHD	0.0	0.0	0.0	0.0	0.0	0.0
PDVD	30.0	30.0	0.0	0.0	0.0	0.0
FDVD	0.0	0.0	0.0	0.0	0.5	2.2
FDHD	0.0	0.0	0.0	0.0	0.0	1.1
ND-SAND	0.0	0.0	0.0	0.0	0.0	15.0
ND-LAr+TMS	0.0	0.0	0.0	0.0	0.0	0.0

Figure 2: Number of raw data events per year, units are Millions.

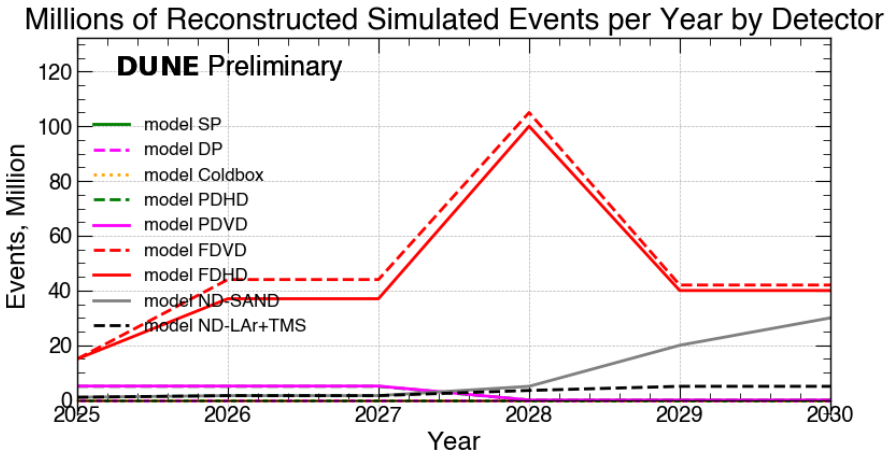
Those data are then stored with sizes/event `RawDataStore` and `SimDataStore`. See Table **??** for storage estimates per event by detector.

3.2 Simulation/Reconstruction Processing

Processing is modeled by taking the number of events produced in a given year and multiplying by the measured CPU times `RecoDataCPU` and `SimDataCPU` for processing those events. Those data are then assumed to be stored with size `RecoDataStore` and `SimDataStore`.

Reprocessing for detector data is assumed to happen every year and cover `Reprocess` $= 3$ years for ProtoDUNE and the whole sample for the FD and ND.

New simulation is assumed to happen every year and previous years are not reprocessed.



Detectors	2025	2026	2027	2028	2029	2030
SP	0.0	0.0	0.0	0.0	0.0	0.0
DP	0.0	0.0	0.0	0.0	0.0	0.0
Coldbox	0.0	0.0	0.0	0.0	0.0	0.0
PDHD	5.0	5.0	5.0	0.0	0.0	0.0
PDVD	5.1	5.1	5.1	0.0	0.0	0.0
FDVD	15.0	44.0	44.0	105.0	42.0	42.0
FDHD	15.0	37.0	37.0	100.0	40.0	40.0
ND-SAND	1.0	1.6	1.6	5.0	20.0	30.0
ND-LAr+TMS	1.0	1.6	1.6	3.5	5.0	5.0

Figure 3: Number of simulated/reconstructed events per year, units are Millions.

3.3 Copies and Retention

Once the data volume generated per year is determined, those data are given retention times and a proposed number of copies. For popular datasets the number of disk copies should be two and disk copies should be retained for 2 years (two reconstruction/simulation cycles) if possible. Due to the large volume of simulation and need for space for raw data from protoDUNE, the number of copies for simulation has been reduced to 1.5 to keep disk demands reasonable.

Raw data has 2 copies on tape with a very long lifetime but a shorter disk lifetime as it is assumed that most users will use the reconstructed samples.

3.4 Analysis

The Analysis model is currently crude. Analysis processing is modeled by scaling the reconstruction and simulation CPU time by a multiplicative factor $\texttt{AnalysisCPU} \sim 0.25 - 0.5$ and extending the time by $\texttt{AnalysisExtend} = 2$ years. Analysis use of reconstructed and simulated data is modeled by extending the disk lifetime of those data by the same amount. Currently analysis data samples are assumed to

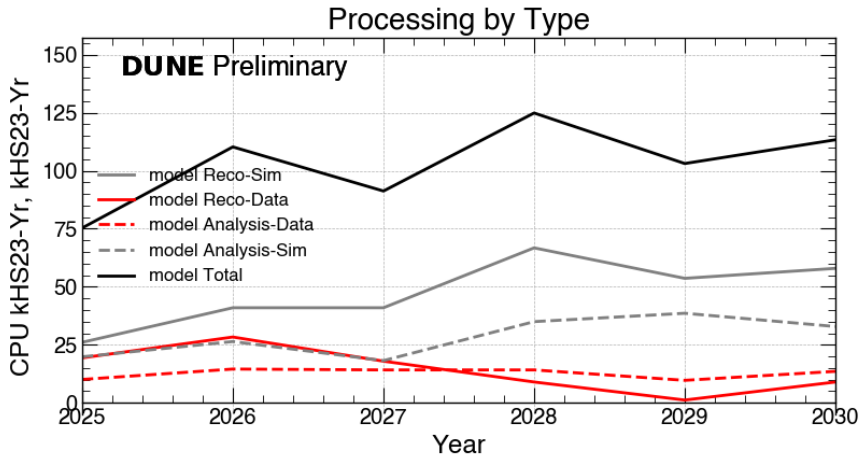
parameter	units	SP	DP	Coldbox	PDHD	PDVD	FDVD	FDHD	ND-SAND	ND-LAr+TMS
Raw-Data-Store	MB	70	110	30.0	140	155	8000	3750	5	10
Reco-Data-Store	MB	35	35	0	35	20	20	20	10	20
Reco-Sim-Store	MB	220	220	0	220	220	20	20	25	419
Reco-Data-CPU	hr	0.1667	0.1667	0.0	0.1667	0.1667	1.33	1.25	0.0022	0.76
Reco-Sim-CPU	hr	0.75	0.75	0.0	0.75	0.75	0.125	0.125	0.24	2.97
Reco-Data-GPU	hr	0	0	0	0	0	0.1	0	0	0.76
Reco-Sim-GPU	hr	0	0	0	0	0	0.1	0	0	0.0222
Analysis-CPU	fraction	0.5	0.5	0.1	0.5	0.5	0.5	0.5	0.25	0.25

Table 2: Per event output size and CPU time parameters used in the model. These values are estimated from running jobs over small numbers of events from protoDUNE and simulation.

Parameters	DiskCopies	DiskLifetimes	TapeCopies	TapeLifetimes
Raw-Data	1	3	2	100
Test	1	0.5	1	1.0
TP	1	0.5	1	100
Reco-Data	2	2	1	15
Reco-Sim	1.5	2	1	15
Analysis-Data	2	5	1	15
Analysis-Sim	2	5	1	15

Table 3: Lifetimes and number of copies for different kinds of data. An exception, we assume protoDUNE raw radata will stay on disk for up to three years for reprocessing. Far detector data are assumed to stay for 2 years.

be much smaller than the reconstructed and simulated samples. Figure/Table **??** and **??**. shows CPU usage by data type.



DataTypes	2025	2026	2027	2028	2029	2030
Reco-Sim	26.1	41.0	41.0	66.8	53.6	57.9
Reco-Data	19.4	28.3	17.9	9.0	1.2	9.0
Analysis-Data	10.1	14.6	14.2	14.2	9.7	13.6
Analysis-Sim	19.8	26.4	18.2	35.0	38.6	32.9
Total	75.3	110.3	91.3	124.9	103.1	113.3

Figure 4: CPU kHS23-Yr resources by data types by year.

3.5 Splitting resources across sites

Resources are assumed to be split as follows:

- Raw data for ProtoDUNE are stored at both CERN and Fermilab
- Reconstructed and simulated data are split between the US (US) and the rest of the Collaboration (Global)

4 Resource use projections

In this section we show projected resource needs by detector, data type, resource type and site.

Detector Class	Data Type	US	CERN	Global
PD	Raw-Data	0.5	0.5	0.0
PD	Reco-Sim	0.25	0.0	0.75
PD	Reco-Data	0.25	0.0	0.75
PD	Test	0.5	0.5	0.0
PD	TP	0.5	0.5	0.0
FD	Raw-Data	0.5	0.0	0.5
FD	Reco-Sim	0.25	0.0	0.75
FD	Reco-Data	0.25	0.0	0.75
FD	Test	0.5	0.0	0.5
FD	TP	0.5	0.0	0.5
ND	Raw-Data	0.5	0.0	0.5
ND	Reco-Sim	0.25	0.0	0.75
ND	Reco-Data	0.25	0.0	0.75
ND	Test	0.5	0.0	0.5
ND	TP	0.5	0.0	0.5

Table 4: Assumptions about splits of Disk resources between the US, CERN and Global.

5 Longer term projections

This section shows the projected resources past the startup of the far detectors in 2028-29.

The estimates for storage are dominated by the far detector while CPU is likely dominated by the near detectors

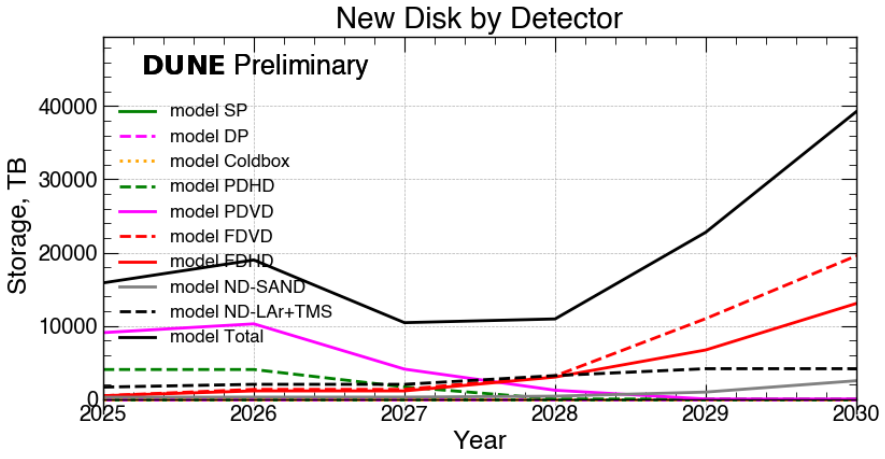
In this section we show projected resource needs by detector, data type, resource type and site.

Detector Class	Data Type	US	CERN	Global
PD	Raw-Data	0.5	0.5	0.0
PD	Reco-Sim	1.0	0.0	0.0
PD	Reco-Data	1.0	0.0	0.0
PD	Test	0.5	0.5	0.0
PD	TP	0.5	0.5	0.0
FD	Raw-Data	0.5	0.0	0.5
FD	Reco-Sim	0.5	0.0	0.5
FD	Reco-Data	0.5	0.0	0.5
FD	Test	0.5	0.0	0.5
FD	TP	0.5	0.0	0.5
ND	Raw-Data	0.5	0.0	0.5
ND	Reco-Sim	0.5	0.0	0.5
ND	Reco-Data	0.5	0.0	0.5
ND	Test	0.5	0.0	0.5

Table 5: Assumptions about splits of Tape resources between the US, CERN and Global.

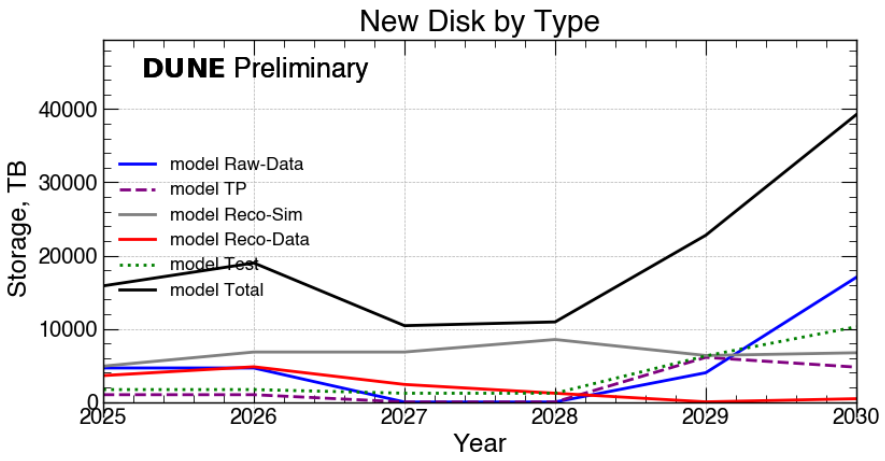
Detector Class	Data Type	US	CERN	Global
PD	CPU	0.4	0.1	0.5
FD	CPU	0.5	0.0	0.5
ND	CPU	0.5	0.0	0.5

Table 6: Assumptions about splits of CPU resources between the US, CERN and Global.



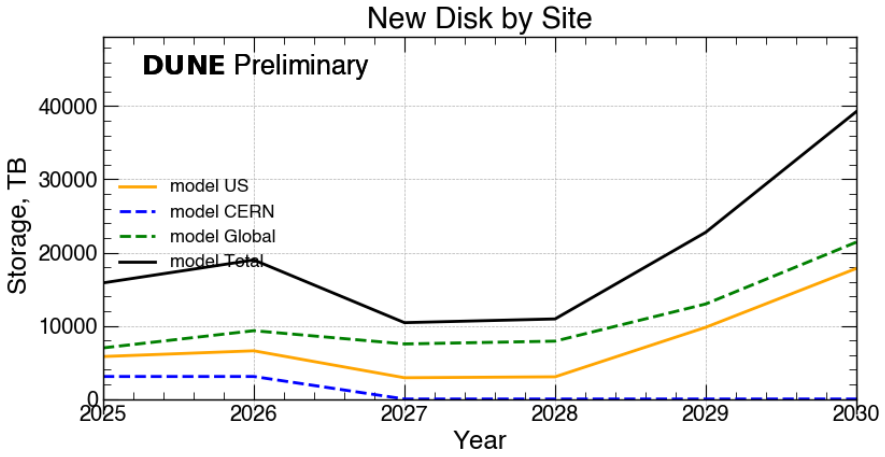
Detectors	2025	2026	2027	2028	2029	2030
SP	0	0	0	0	0	0
DP	0	0	0	0	0	0
Coldbox	0	0	0	0	0	0
PDHD	4030	4030	1650	0	0	0
PDVD	9065	10265	4083	1200	0	0
FDVD	450	1320	1320	3150	10980	19559
FDHD	450	1110	1110	3000	6700	13052
ND-SAND	237	260	260	387	950	2500
ND-LAr+TMS	1628	2005	2005	3199	4142	4142
Total	15861	18990	10428	10937	22772	39254

Figure 5: New Disk per year by detector. This includes multiple copies where need but not longer lifetimes.



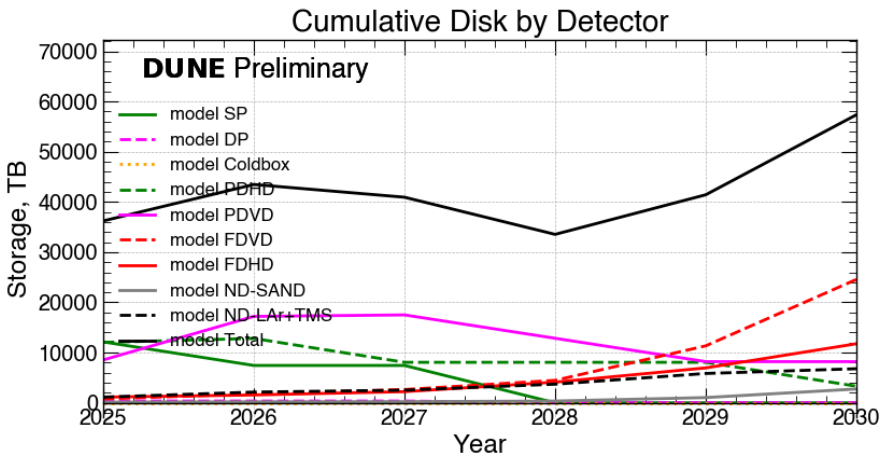
DataTypes	2025	2026	2027	2028	2029	2030
Raw-Data	4650	4650	0	0	4000	17036
TP	1000	1000	0	0	6100	4762
Reco-Sim	4899	6828	6828	8537	6352	6727
Reco-Data	3612	4812	2400	1200	20	452
Test	1700	1700	1200	1200	6300	10275
Total	15861	18990	10428	10937	22772	39254

Figure 6: New Disk per year by data type. This includes multiple copies where need but not longer lifetimes.



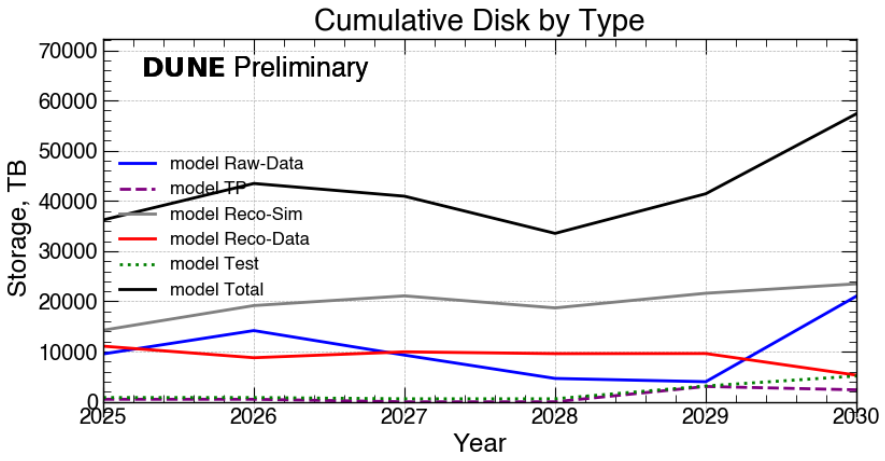
Locations	2025	2026	2027	2028	2029	2030
US	5802	6585	2907	3034	9793	17832
CERN	3075	3075	0	0	0	0
Global	6983	9330	7521	7902	12979	21422
Total	15861	18990	10428	10937	22772	39254

Figure 7: New Disk per year by site. This includes multiple copies where need but not longer lifetimes.



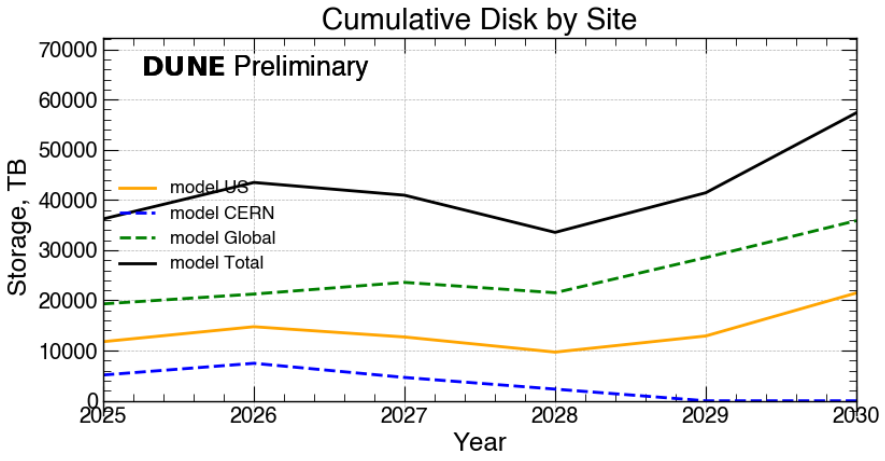
Detectors	2025	2026	2027	2028	2029	2030
SP	12132	7435	7435	0	0	0
DP	355	355	355	0	0	0
Coldbox	0	0	0	0	0	0
PDHD	12028	12820	8060	8060	8060	3300
PDVD	8532	17204	17498	12848	8198	8198
FDVD	775	1770	2640	4470	11340	24564
FDHD	1128	1560	2220	4110	6950	11758
ND-SAND	137	197	220	347	1037	2750
ND-LAr+TMS	1128	2134	2511	3705	5842	6785
Total	36218	43476	40940	33540	41427	57355

Figure 8: Cumulative Disk by detector.



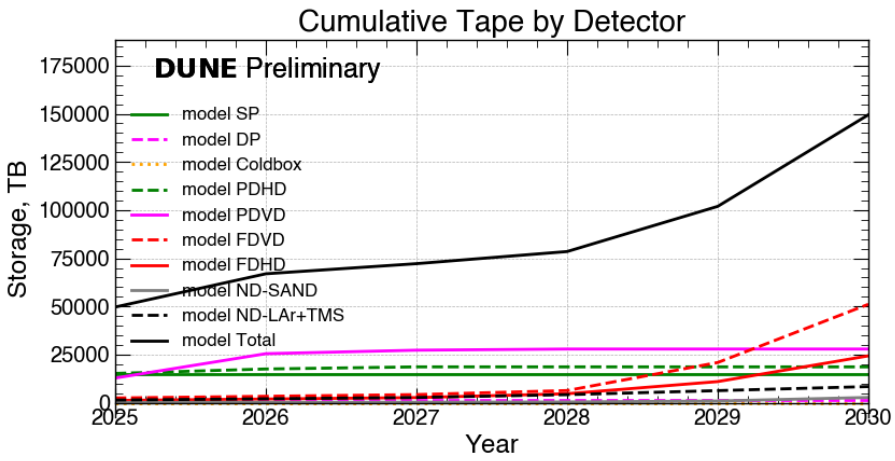
DataTypes	2025	2026	2027	2028	2029	2030
Raw-Data	9534	14184	9300	4650	4000	21036
TP	500	500	0	0	3050	2381
Reco-Sim	14258	19163	21093	18698	21615	23496
Reco-Data	11076	8779	9947	9592	9612	5304
Test	850	850	600	600	3150	5137
Total	36218	43476	40940	33540	41427	57355

Figure 9: Cumulative Disk by data type.



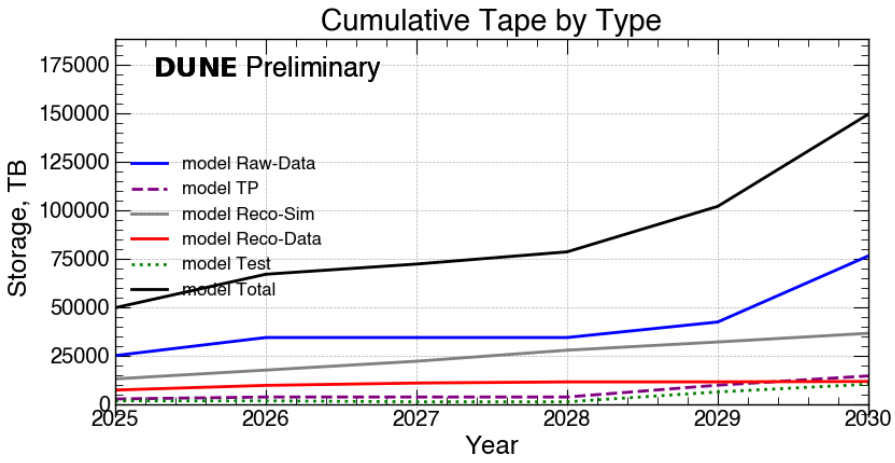
Locations	2025	2026	2027	2028	2029	2030
US	11775	14752	12710	9697	12906	21477
CERN	5142	7467	4650	2325	0	0
Global	19300	21256	23580	21518	28520	35877
Total	36218	43476	40940	33540	41427	57355

Figure 10: Cumulative Disk by site.



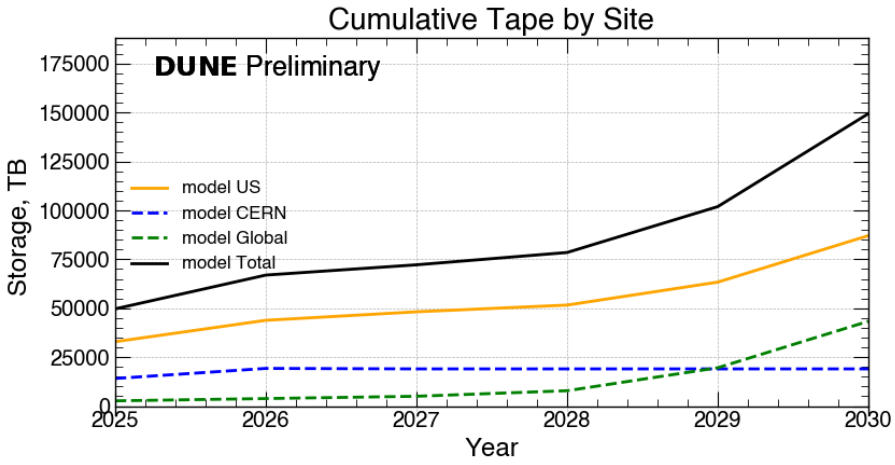
Detectors	2025	2026	2027	2028	2029	2030
SP	15021	15021	15021	15021	15021	15021
DP	1208	1208	1208	1208	1208	1208
Coldbox	0	0	0	0	0	0
PDHD	15251	17541	18641	18641	18641	18641
PDVD	12843	25481	27303	27903	27903	27903
FDVD	2500	3380	4260	6360	20910	51136
FDHD	1224	1964	2704	4704	11004	24355
ND-SAND	225	265	305	430	930	2780
ND-LAr+TMS	1419	2089	2759	4226	6321	8416
Total	49693	66951	72203	78495	101940	149462

Figure 11: Cumulative Tape by detector.



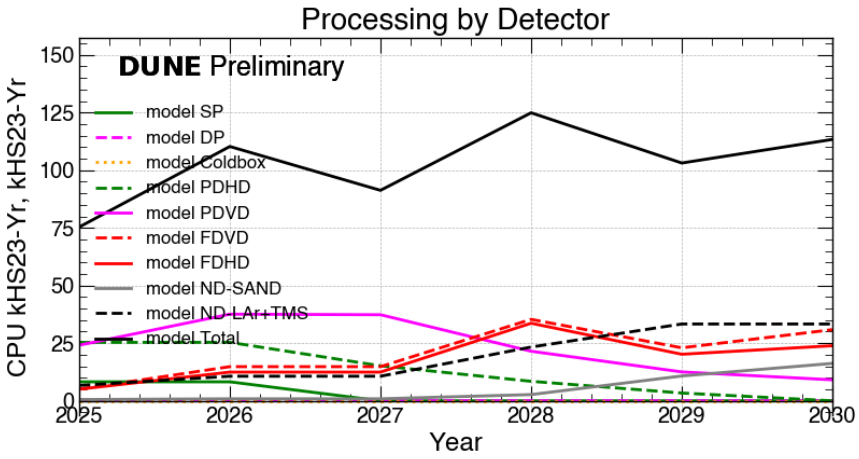
DataTypes	2025	2026	2027	2028	2029	2030
Raw-Data	25038	34338	34338	34338	42338	76411
TP	2679	3679	3679	3679	9779	14541
Reco-Sim	13009	17561	22113	27805	32040	36525
Reco-Data	7266	9672	10872	11472	11482	11708
Test	1700	1700	1200	1200	6300	10275
Total	49693	66951	72203	78495	101940	149462

Figure 12: Cumulative Tape by data type.



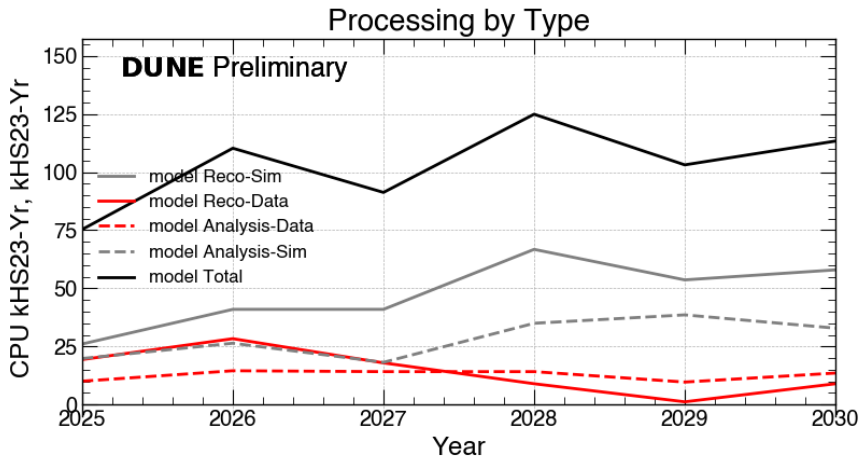
Locations	2025	2026	2027	2028	2029	2030
US	32900	43843	48180	51626	63348	87110
CERN	14108	19258	19008	19008	19008	19008
Global	2684	3849	5014	7860	19582	43344
Total	49693	66951	72203	78495	101940	149462

Figure 13: Cumulative Tape by site.



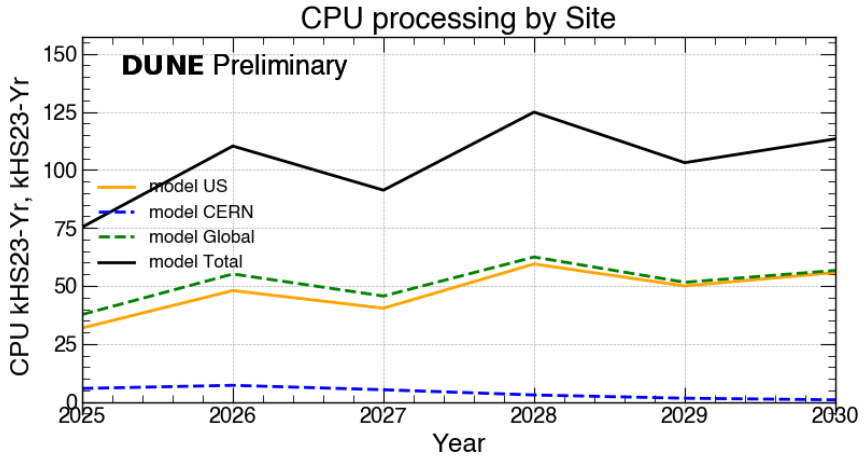
Detectors	2025	2026	2027	2028	2029	2030
SP	8.2	8.2	0.0	0.0	0.0	0.0
DP	0.4	0.4	0.0	0.0	0.0	0.0
Coldbox	0.0	0.0	0.0	0.0	0.0	0.0
PDHD	25.3	25.3	15.2	8.4	3.4	0.0
PDVD	24.1	37.6	37.3	21.5	12.5	9.1
FDVD	5.0	14.8	14.8	35.3	23.0	30.9
FDHD	5.0	12.4	12.4	33.6	20.2	23.9
ND-SAND	0.5	0.9	0.9	2.7	10.8	16.2
ND-LAr+TMS	6.7	10.7	10.7	23.3	33.3	33.3
Total	75.3	110.3	91.3	124.9	103.1	113.3

Figure 14: CPU kHS23-Yr resources by detector by year.



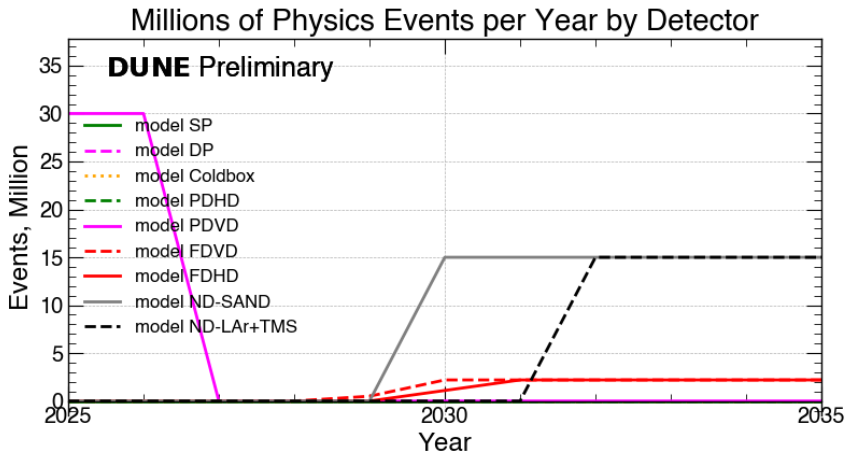
DataTypes	2025	2026	2027	2028	2029	2030
Reco-Sim	26.1	41.0	41.0	66.8	53.6	57.9
Reco-Data	19.4	28.3	17.9	9.0	1.2	9.0
Analysis-Data	10.1	14.6	14.2	14.2	9.7	13.6
Analysis-Sim	19.8	26.4	18.2	35.0	38.6	32.9
Total	75.3	110.3	91.3	124.9	103.1	113.3

Figure 15: CPU kHS23-Yr resources by data types by year.



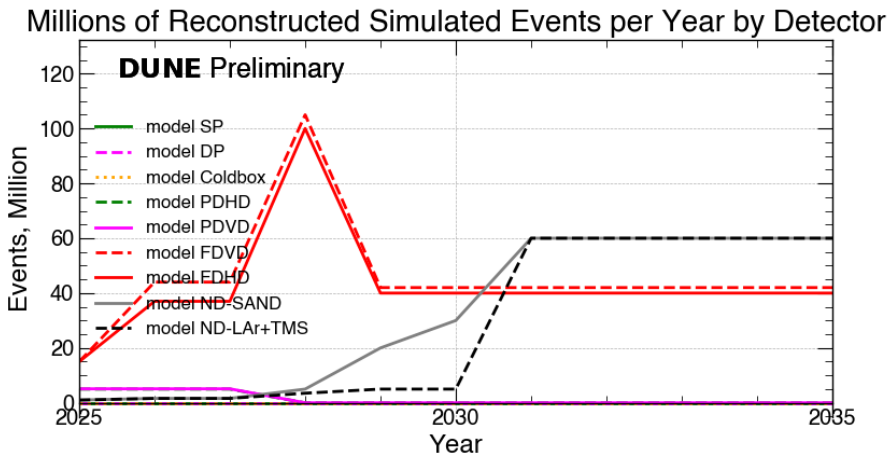
Locations	2025	2026	2027	2028	2029	2030
US	31.9	48.0	40.4	59.5	50.0	55.8
CERN	5.8	7.2	5.2	3.0	1.6	0.9
Global	37.7	55.1	45.6	62.4	51.6	56.7
Total	75.3	110.3	91.3	124.9	103.1	113.3

Figure 16: Projected CPU use by location. Units are CPU kHS23-Yr



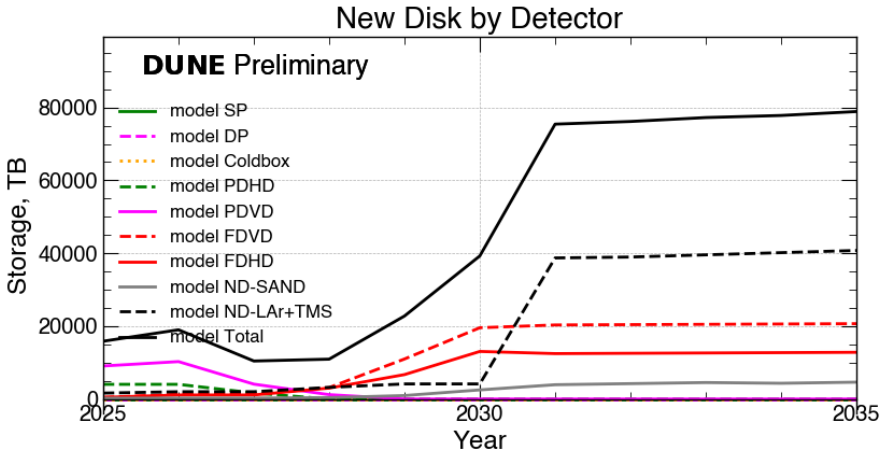
Detectors	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
SP	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
DP	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Coldbox	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
PDHD	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
PDVD	30.0	30.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
FDVD	0.0	0.0	0.0	0.0	0.5	2.2	2.2	2.2	2.2	2.2	2.2
FDHD	0.0	0.0	0.0	0.0	0.0	1.1	2.2	2.2	2.2	2.2	2.2
ND-SAND	0.0	0.0	0.0	0.0	0.0	15.0	15.0	15.0	15.0	15.0	15.0
ND-LAr+TMS	0.0	0.0	0.0	0.0	0.0	0.0	0.0	15.0	15.0	15.0	15.0

Figure 17: Number of raw data events per year, units are Millions.



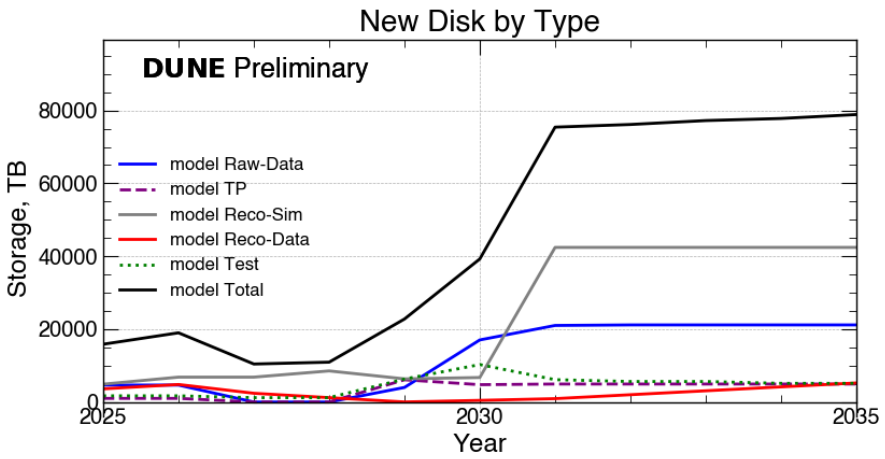
Detectors	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
SP	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
DP	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Coldbox	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
PDHD	5.0	5.0	5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
PDVD	5.1	5.1	5.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
FDVD	15.0	44.0	44.0	105.0	42.0	42.0	42.0	42.0	42.0	42.0	42.0
FDHD	15.0	37.0	37.0	100.0	40.0	40.0	40.0	40.0	40.0	40.0	40.0
ND-SAND	1.0	1.6	1.6	5.0	20.0	30.0	60.0	60.0	60.0	60.0	60.0
ND-LAr+TMS	1.0	1.6	1.6	3.5	5.0	5.0	60.0	60.0	60.0	60.0	60.0

Figure 18: Number of simulated/reconstructed events per year, units are Millions.



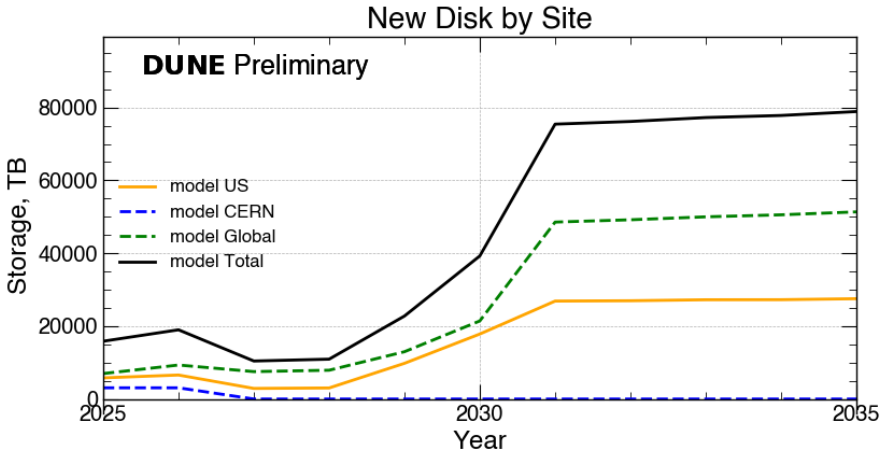
Detectors	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
SP	0	0	0	0	0	0	0	0	0	0	0
DP	0	0	0	0	0	0	0	0	0	0	0
Coldbox	0	0	0	0	0	0	0	0	0	0	0
PDHD	4030	4030	1650	0	0	0	0	0	0	0	0
PDVD	9065	10265	4083	1200	0	0	0	0	0	0	0
FDVD	450	1320	1320	3150	10980	19559	20322	20410	20498	20586	20674
FDHD	450	1110	1110	3000	6700	13052	12465	12553	12641	12729	12817
ND-SAND	237	260	260	387	950	2500	3925	4225	4525	4325	4625
ND-LAr+TMS	1628	2005	2005	3199	4142	4142	38710	38960	39560	40160	40760
Total	15861	18990	10428	10937	22772	39254	75423	76149	77225	77801	78877

Figure 19: New Disk per year by detector. This includes multiple copies where need but not longer lifetimes.



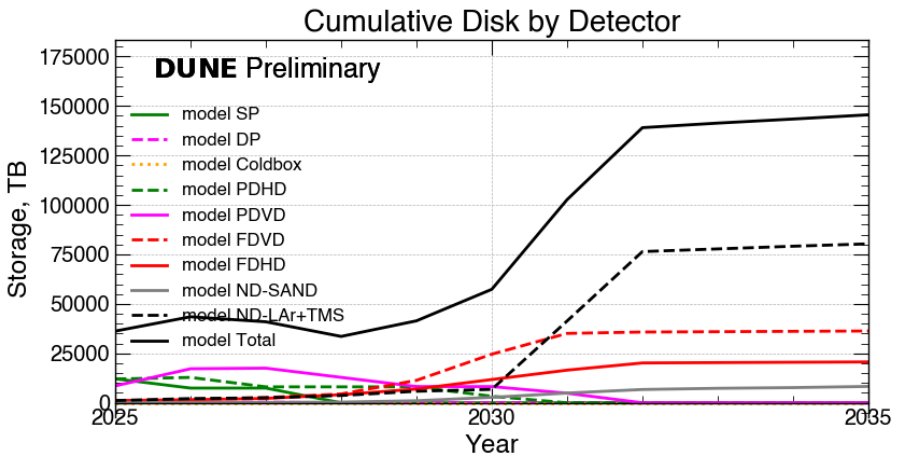
DataTypes	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
Raw-Data	4650	4650	0	0	4000	17036	21006	21156	21156	21156	21156
TP	1000	1000	0	0	6100	4762	4939	4939	4939	4939	4939
Reco-Sim	4899	6828	6828	8537	6352	6727	42420	42420	42420	42420	42420
Reco-Data	3612	4812	2400	1200	20	452	928	2004	3080	4156	5232
Test	1700	1700	1200	1200	6300	10275	6129	5629	5629	5129	5129
Total	15861	18990	10428	10937	22772	39254	75423	76149	77225	77801	78877

Figure 20: New Disk per year by data type. This includes multiple copies where need but not longer lifetimes.



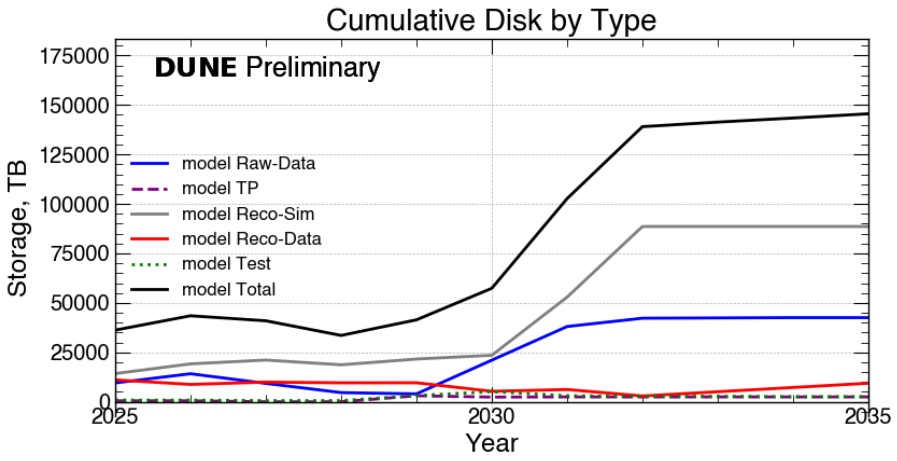
Locations	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
US	5802	6585	2907	3034	9793	17832	26874	26968	27237	27256	27525
CERN	3075	3075	0	0	0	0	0	0	0	0	0
Global	6983	9330	7521	7902	12979	21422	48548	49180	49987	50544	51351
Total	15861	18990	10428	10937	22772	39254	75423	76149	77225	77801	78877

Figure 21: New Disk per year by site. This includes multiple copies where need but not longer lifetimes.



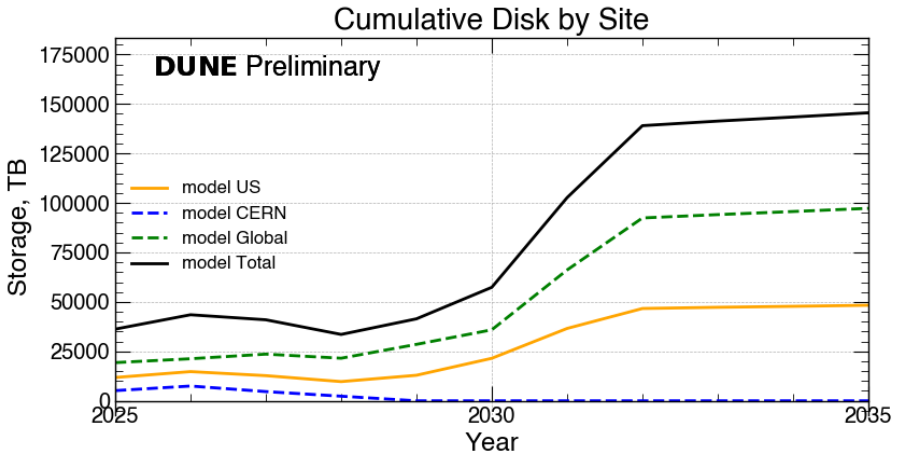
Detectors	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
SP	12132	7435	7435	0	0	0	0	0	0	0	0
DP	355	355	355	0	0	0	0	0	0	0	0
Coldbox	0	0	0	0	0	0	0	0	0	0	0
PDHD	12028	12820	8060	8060	8060	3300	0	0	0	0	0
PDVD	8532	17204	17498	12848	8198	8198	4832	0	0	0	0
FDVD	775	1770	2640	4470	11340	24564	35073	35759	35935	36111	36287
FDHD	1128	1560	2220	4110	6950	11758	16503	20139	20315	20491	20667
ND-SAND	137	197	220	347	1037	2750	4925	6725	7325	7675	8275
ND-LAr+TMS	1128	2134	2511	3705	5842	6785	41352	76420	77770	79120	80320
Total	36218	43476	40940	33540	41427	57355	102686	139043	141345	143397	145549

Figure 22: Cumulative Disk by detector.



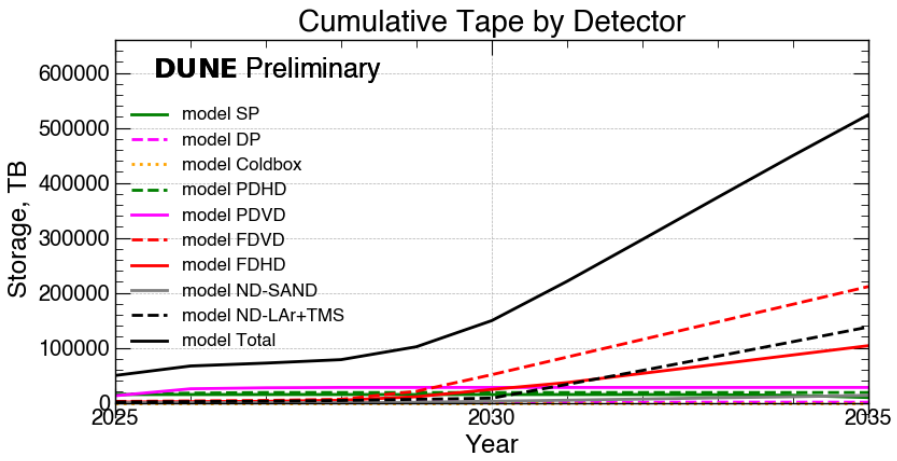
DataTypes	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
Raw-Data	9534	14184	9300	4650	4000	21036	38042	42237	42387	42537	42537
TP	500	500	0	0	3050	2381	2469	2469	2469	2469	2469
Reco-Sim	14258	19163	21093	18698	21615	23496	52897	88590	88590	88590	88590
Reco-Data	11076	8779	9947	9592	9612	5304	6212	2932	5084	7236	9388
Test	850	850	600	600	3150	5137	3064	2814	2814	2564	2564
Total	36218	43476	40940	33540	41427	57355	102686	139043	141345	143397	145549

Figure 23: Cumulative Disk by data type.



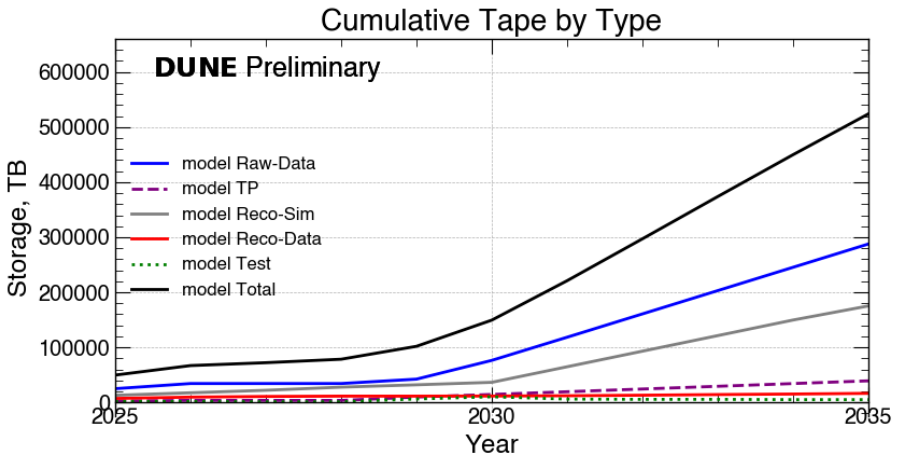
Locations	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
US	11775	14752	12710	9697	12906	21477	36565	46641	47254	47742	48280
CERN	5142	7467	4650	2325	0	0	0	0	0	0	0
Global	19300	21256	23580	21518	28520	35877	66120	92402	94091	95655	97269
Total	36218	43476	40940	33540	41427	57355	102686	139043	141345	143397	145549

Figure 24: Cumulative Disk by site.



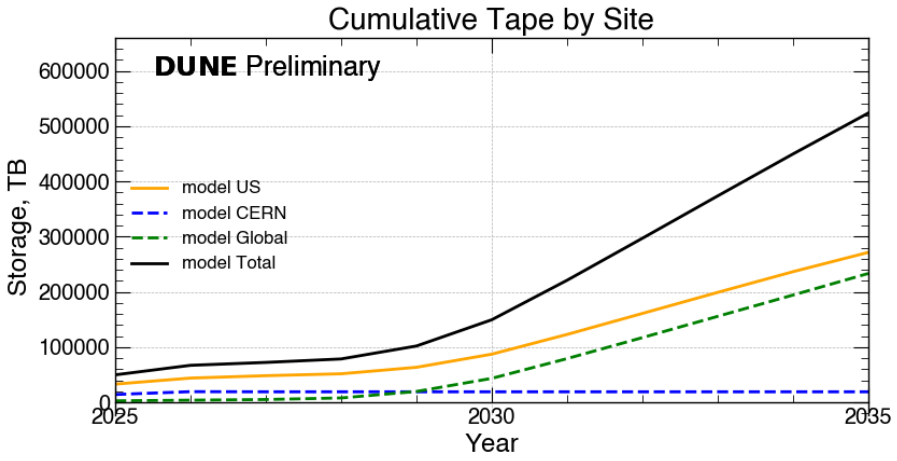
Detectors	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
SP	15021	15021	15021	15021	15021	15021	15021	15021	14640	13579	9600
DP	1208	1208	1208	1208	1208	1208	1208	1208	1208	1152	1063
Coldbox	0	0	0	0	0	0	0	0	0	0	0
PDHD	15251	17541	18641	18641	18641	18641	18641	18641	18641	18641	18641
PDVD	12843	25481	27303	27903	27903	27903	27903	27903	27903	27903	27903
FDVD	2500	3380	4260	6360	20910	51136	83161	115156	147194	179276	211402
FDHD	1224	1964	2704	4704	11004	24355	36789	53488	70232	87019	103851
ND-SAND	225	265	305	430	930	2780	4730	6830	9080	10980	13530
ND-LAr+TMS	1419	2089	2759	4226	6321	8416	33556	58796	84836	111176	137816
Total	49693	66951	72203	78495	101940	149462	221012	297045	373736	449729	523808

Figure 25: Cumulative Tape by detector.



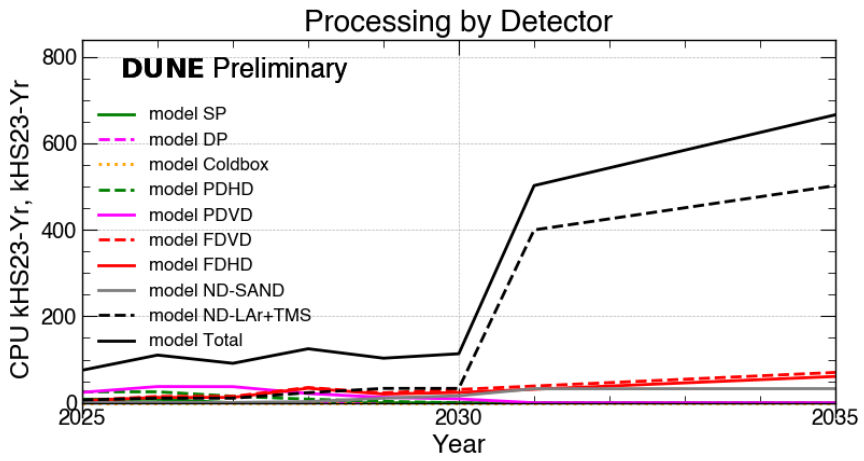
DataTypes	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
Raw-Data	25038	34338	34338	34338	42338	76411	118423	160736	203048	245360	287673
TP	2679	3679	3679	3679	9779	14541	19481	24420	29359	34298	39238
Reco-Sim	13009	17561	22113	27805	32040	36525	64805	93085	121365	149645	175234
Reco-Data	7266	9672	10872	11472	11482	11708	12172	13174	14332	15294	16533
Test	1700	1700	1200	1200	6300	10275	6129	5629	5629	5129	5129
Total	49693	66951	72203	78495	101940	149462	221012	297045	373736	449729	523808

Figure 26: Cumulative Tape by data type.



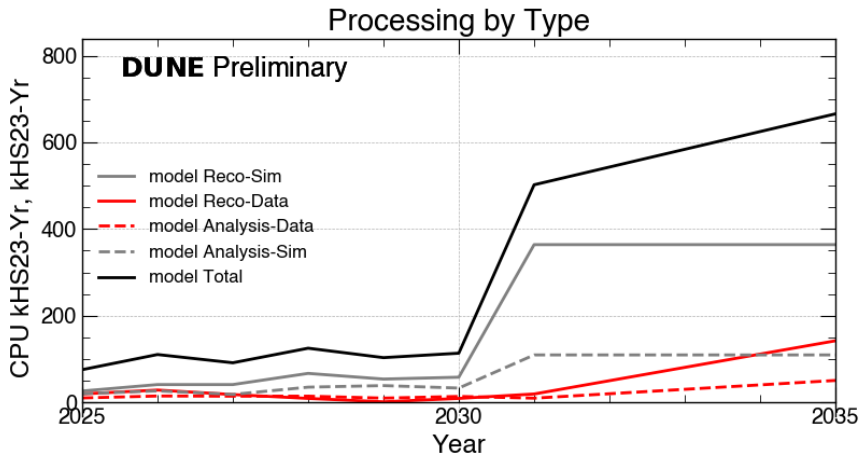
Locations	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
US	32900	43843	48180	51626	63348	87110	122884	160901	199055	236494	271499
CERN	14108	19258	19008	19008	19008	19008	19008	19008	19008	19008	19008
Global	2684	3849	5014	7860	19582	43344	79118	117135	155671	194226	233300
Total	49693	66951	72203	78495	101940	149462	221012	297045	373736	449729	523808

Figure 27: Cumulative Tape by site.



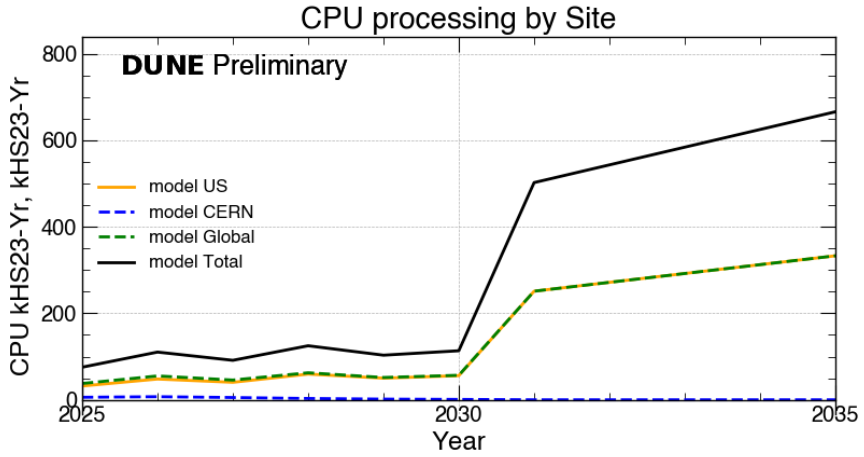
Detectors	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
SP	8.2	8.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
DP	0.4	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Coldbox	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
PDHD	25.3	25.3	15.2	8.4	3.4	0.0	0.0	0.0	0.0	0.0	0.0
PDVD	24.1	37.6	37.3	21.5	12.5	9.1	0.0	0.0	0.0	0.0	0.0
FDVD	5.0	14.8	14.8	35.3	23.0	30.9	38.7	46.6	54.5	62.3	70.2
FDHD	5.0	12.4	12.4	33.6	20.2	23.9	31.3	38.7	46.1	53.5	60.9
ND-SAND	0.5	0.9	0.9	2.7	10.8	16.2	32.4	32.5	32.6	32.7	32.7
ND-LAr+TMS	6.7	10.7	10.7	23.3	33.3	33.3	399.6	425.1	450.7	476.3	501.8
Total	75.3	110.3	91.3	124.9	103.1	113.3	502.0	542.9	583.8	624.8	665.7

Figure 28: CPU kHS23-Yr resources by detector by year.



DataTypes	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
Reco-Sim	26.1	41.0	41.0	66.8	53.6	57.9	363.9	363.9	363.9	363.9	363.9
Reco-Data	19.4	28.3	17.9	9.0	1.2	9.0	19.2	49.9	80.6	111.3	142.0
Analysis-Data	10.1	14.6	14.2	14.2	9.7	13.6	9.6	19.8	30.0	40.2	50.4
Analysis-Sim	19.8	26.4	18.2	35.0	38.6	32.9	109.4	109.4	109.4	109.4	109.4
Total	75.3	110.3	91.3	124.9	103.1	113.3	502.0	542.9	583.8	624.8	665.7

Figure 29: CPU kHS23-Yr resources by data types by year.



Locations	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
US	31.9	48.0	40.4	59.5	50.0	55.8	251.0	271.5	291.9	312.4	332.8
CERN	5.8	7.2	5.2	3.0	1.6	0.9	0.0	0.0	0.0	0.0	0.0
Global	37.7	55.1	45.6	62.4	51.6	56.7	251.0	271.5	291.9	312.4	332.8
Total	75.3	110.3	91.3	124.9	103.1	113.3	502.0	542.9	583.8	624.8	665.7

Figure 30: Projected CPU use by location. Units are CPU kHS23-Yr

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[3] H. Schellman, "DUNE Offline Computing Model Calculations for 2024," https://docs.dunescience.org/cgi-bin/sso/ShowDocument?docid=29925.

[4] P. Clarke, "Computing Contribution Board (CCB) Minutes for February 2024," https://docs.dunescience.org/cgi-bin/sso/ShowDocument?docid=30302.